42 Tokamac

'Tokamac' is the name for a magnetic field system used with nuclear fusion in the JET undertaking. (See question 41.) Part of the action is that of a transformer, with the plasma in its toroidal vacuum chamber acting as the secondary coil. A huge current, of up to seven million amps, can be induced in the plasma for a short time, and it is hoped that this will heat up the plasma to a temperature where the fusion reaction becomes self sustaining, although other methods of heating, such as injecting energetic particles into the plasma, and the use of radio beams, are also employed. The layout of the Tokamac is shown in the diagram below, and the graphs show the currents that have been achieved in the plasma.

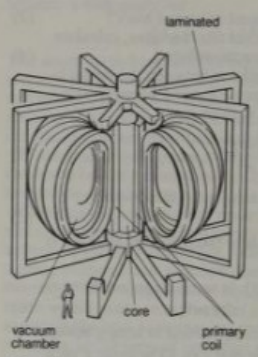


Figure 1

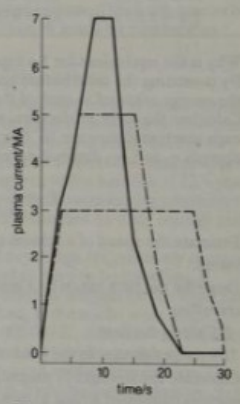


Figure 2

The large amount of electrical energy needed to produce the plasma current cannot be taken directly from the National Grid, even over the short time needed, as the power required is in excess of 700 MW. One method of storing the energy would be to use a large bank of capacitors, which could be charged up individually, then connected in parallel, and all discharged through the primary coil at the same time.

78

The method which is actually used in the JET programme, however, is to have two flywheel generators. Each one has a 9.0 m diameter rotor with a mass of 780 tonnes. The speed of the rotor is gradually built up to 230 rpm using an 8.8 MW motor. When a pulse of current is needed, the generator coils are connected, and each rotor slows down to half speed in a short time. If we assume that the flywheels are simple discs, the amounts of energy involved can be calculated; the rotational kinetic energy (in J) of a disc is given by $Mr^2\omega^2/4$, where M is the mass (in kg), r is the radius (in m) and ω is the angular speed (in radians/s).

(a) Explain the principle of the transformer action of the Tokamac; how can a large current be induced in the plasma by passing a much smaller current through the primary coil? *(4)*

(b) Most commercial transformers have two branches for the magnetic circuit of the iron. Why does the Tokamac have several? *(2)*

(c) Estimate the total charge that passes through the plasma as a result of the 7.0 MA current, and explain how you made your estimate. *(3)*

(d) If capacitors were used to provide this charge directly, and each one had a safe working voltage of 100 V, calculate the total capacitance needed and the energy stored. *(4)*

(e) The capacitance of a parallel plate capacitor is given by

$$C = \frac{\varepsilon_o \varepsilon_r A}{d}$$

If ε_o is $8.9 \times 10^{-12}\,\mathrm{Fm^{-1}}$, ε_r is 3.0, and d is 0.10 mm, calculate the area A of the plates needed, and estimate the volume of capacitors needed. Why do you think the flywheel generator option was chosen? *(5)*

(f) Estimate the time for which the maximum current of 7.0 MA passes through the plasma, and use this with the peak power stated in the passage to calculate the energy supplied. Is the energy you calculated in question (d) of the right order of magnitude? *(4)*

(g) Calculate the rotational kinetic energy of one flywheel generator when spinning at its full speed, and when slowed down to half speed (1 revolution is 2π radians). Hence calculate the energy available from both generators to produce the pulse of current. *(5)*

(h) How long will the two 8.8 MW motors take, in theory, to bring the flywheels back to their maximum speed? Why will the actual time be longer than this? *(3)*

Total 30 marks

79

CIE 9702 Physics P4 Style Question — Tokamak and Pulse Heating

Read the following passage carefully before answering the questions.

The term *Tokamak* refers to a magnetic confinement system used in fusion research. In such a device, a toroidal plasma is confined by magnetic fields produced by large current-carrying coils. In one method of heating, the plasma itself acts as the secondary winding of a transformer. A changing current in the primary coils induces an emf in the plasma ring, driving a very large plasma current for a short duration. This induced current causes Joule heating and helps raise the plasma temperature.

For a large experimental system, direct energy supply from the electrical grid is limited. A pulse-heating method is therefore used: two heavy flywheel generators are first spun up by motors, then clutched to alternators so that their rotational kinetic energy can be discharged rapidly into the primary coils.

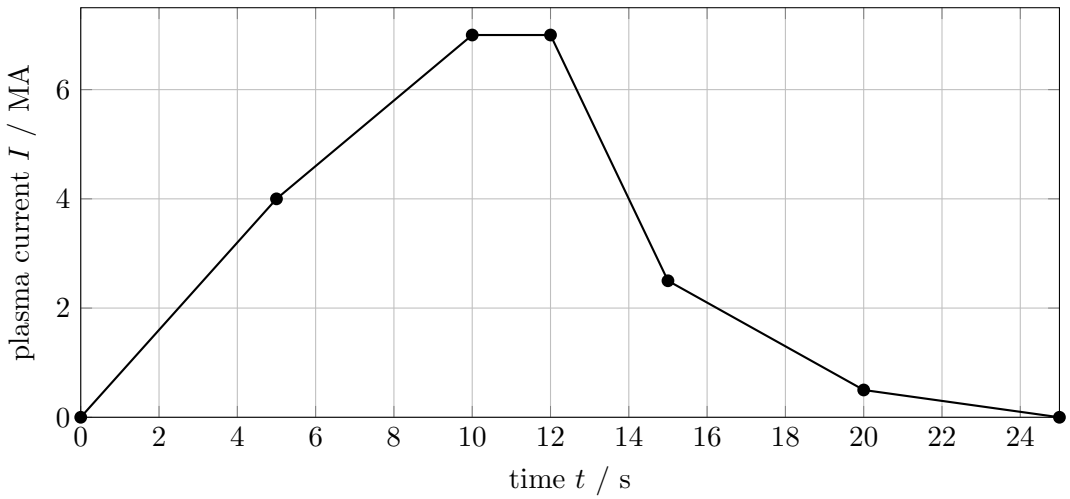
Each flywheel is modeled as a solid disc of mass $m = 780\,\text{tonnes} = 7.80 \times 10^5\,\text{kg}$ and radius $r = 5.0\,\text{m}$. At maximum speed, the flywheel rotates at $230\,\text{rev min}^{-1}$. The total instantaneous electrical power available to the pulse system is approximately 700 MW.

The target plasma current pulse rises rapidly to a maximum and then decays. A simplified set of measured values for one pulse is shown in Table 1.

Table 1: Plasma current during one pulse

t/s	0	5	10	12	15	20	25
I/MA	0.0	4.0	7.0	7.0	2.5	0.5	0.0

Figure 1 shows these data.



You may assume that:

$$E_{\text{rot}} = \frac{1}{2} I_{\text{disc}} \omega^2, \qquad I_{\text{disc}} = \frac{1}{2} m r^2, \qquad \omega = 2\pi f.$$

(a) Explain why transformer action can induce a current in the plasma ring, and why this method does *not* provide continuous heating. [4]

(b) Explain why very large currents are needed in the Tokamak primary windings compared with ordinary commercial transformers. [3]

(c) Using Table 1 or Figure 1, estimate the total charge that flows in the plasma during the interval $0 \leq t \leq 25\,\text{s}$. Show your method clearly. [4]

(d) The pulse system uses parallel-plate capacitors rated at 100 V each. Assuming an available pulse energy of 1.4×10^{10} J from the two flywheel systems together, calculate the total capacitance required. [4]

(e) For a parallel-plate capacitor,

$$C = \varepsilon_0 \varepsilon_r \frac{A}{d}.$$

Given $\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$, $\varepsilon_r = 3.0$, and $d = 0.10\,\text{mm}$, estimate the total plate area required from your answer to (d). State one practical reason why a flywheel-assisted pulse system may be preferred over storing all energy electrostatically. [5]

(f) A student approximates the pulse duration as 25 s at a constant current of 7.0 MA. Comment on whether this is a good estimate and explain, using the graph, how this approximation affects the calculated charge and energy transfer. [3]

(g) Calculate the rotational kinetic energy of one flywheel at 230 rev min^{-1}, and hence estimate the total energy from two flywheels. Compare your result with the 1.4×10^{10} J value used earlier. [4]

(h) If the motors that spin up the flywheels can each supply 8.8 MW, estimate the minimum time needed to restore both flywheels to full speed after a pulse. Give one reason why the actual recovery time is longer. [3]

(i) Open-ended: Evaluate the statement: *"Pulsed current heating is an effective but temporary solution for achieving fusion conditions."* In your answer, refer to at least **two** quantitative values from the passage and discuss both advantages and limitations. [6]

Total: 36 marks

Suggested Answers / Marking Points

(a) Current in primary varies with time $\Rightarrow$ changing magnetic flux through plasma loop. By Faraday's law an emf is induced around plasma, so plasma current flows. Continuous DC in primary gives no changing flux, so induced emf falls to zero; therefore this method is inherently pulsed.

(b) Plasma is a single-turn, low-resistance secondary with huge current demand; strong magnetic fields and large induced emf are required in large-volume plasma; coupling is poorer than in iron-core transformers.

(c) Charge $Q = \int I\,dt \approx$ area under I–t graph. Using trapezoids (in MA·s):

$$\tfrac{1}{2}(0+4)(5) + \tfrac{1}{2}(4+7)(5) + \tfrac{1}{2}(7+7)(2) + \tfrac{1}{2}(7+2.5)(3) + \tfrac{1}{2}(2.5+0.5)(5) + \tfrac{1}{2}(0.5+0)(5) = 124.25.$$

Hence $Q \approx 124.25 \times 10^6\,\mathrm{C} \approx 1.24 \times 10^8\,\mathrm{C}$.

(d)

$$E = \tfrac{1}{2}CV^2 \Rightarrow C = \frac{2E}{V^2} = \frac{2(1.4 \times 10^{10})}{(100)^2} = 2.8 \times 10^6\,\mathrm{F}.$$

(e)

$$A = \frac{Cd}{\varepsilon_0 \varepsilon_r} = \frac{(2.8 \times 10^6)(1.0 \times 10^{-4})}{(8.85 \times 10^{-12})(3.0)} \approx 1.05 \times 10^{13}\,\mathrm{m}^2.$$

Area is impractically large. One valid reason: flywheels provide high-energy storage with manageable physical size and lower capacitor bank complexity.

(f) Constant-7.0 MA for 25 s gives $Q = 175 \times 10^6\,\mathrm{C}$, which overestimates compared with $\approx 124 \times 10^6\,\mathrm{C}$. Graph shows maximum current is brief and decay is substantial, so rectangular approximation is poor.

(g)

$$\omega = 2\pi\left(\frac{230}{60}\right) = 24.1\,\mathrm{rad\ s^{-1}}, \quad I_{\mathrm{disc}} = \tfrac{1}{2}mr^2 = \tfrac{1}{2}(7.80 \times 10^5)(5.0)^2 = 9.75 \times 10^6\,\mathrm{kg\ m^2}.$$

$$E_{\mathrm{one}} = \tfrac{1}{2}I\omega^2 \approx \tfrac{1}{2}(9.75 \times 10^6)(24.1)^2 \approx 2.83 \times 10^9\,\mathrm{J}.$$

Two flywheels: $\approx 5.66 \times 10^9$ J. This is lower than 1.4×10^{10} J, suggesting either additional generators/storage stages or idealisations in the model/data.

(h) Total motor power = 17.6 MW. For restoring 5.66×10^9 J,

$$t_{\min} = \frac{E}{P} = \frac{5.66 \times 10^9}{17.6 \times 10^6} \approx 3.2 \times 10^2\,\mathrm{s} \approx 5.4\,\mathrm{min}.$$

Actual time longer due to losses (friction, electrical inefficiency, clutch/generator losses, non-constant motor output).

(i) High-level points for full-credit responses: quantitative reference to peak current (7.0 MA), pulse duration ($\sim$ 25 s), large power (700 MW), and/or flywheel energy scale (10^9–10^{10} J); clear discussion that pulsing can achieve strong heating but is not steady-state and requires recharge time and complex infrastructure.

Variant 2 (Higher Difficulty) — Tokamak Pulse Optimisation and Engineering Constraints

Read the following passage carefully before answering the questions.

A research team upgrades the pulse-heating stage of a Tokamak to improve plasma performance. Two flywheel-generator sets are still used, but the control strategy is modified so that the primary-coil drive current is shaped in time to maintain a high plasma current for longer. The design team must balance three competing goals: high peak current, long pulse duration, and manageable mechanical and electrical stress.

For this upgraded system, each flywheel remains a solid disc with

$$m = 7.80 \times 10^5\,\text{kg}, \qquad r = 5.0\,\text{m}.$$

The maximum permitted rotational speed is increased to $260\,\text{rev min}^{-1}$, but operation above $240\,\text{rev min}^{-1}$ causes rapidly rising bearing losses. The combined motor input power available between pulses is 18.0 MW. During discharge, the electrical conversion efficiency from flywheel rotational energy to useful primary-coil energy is estimated to be 78%.

For one high-performance pulse, the measured plasma current data are shown below.

Table 2: Plasma current for upgraded pulse

t/s	0	4	8	12	16	20	24	28
I/MA	0.0	5.0	8.2	8.0	6.3	3.8	1.2	0.0

The plasma effective resistance during this pulse may be taken as constant at

$$R_{\text{plasma}} = 1.6 \times 10^{-6}\,\Omega.$$

You may assume:

$$E_{\text{rot}} = \frac{1}{2} I_{\text{disc}} \omega^2, \qquad I_{\text{disc}} = \frac{1}{2} m r^2, \qquad P = I^2 R, \qquad E = \int P\,dt.$$

(a) State why increasing flywheel angular speed is an effective way to increase available pulse energy. Explain why this cannot be increased indefinitely in practice. [3]

(b) Using Table 2, estimate the total charge transferred in the plasma during the pulse. [4]

(c) Estimate the total thermal energy dissipated in the plasma resistance during the pulse using numerical integration of I^2R. Show your working clearly. [5]

(d) Determine the rotational kinetic energy stored in one flywheel at:

(i) $240\,\text{rev min}^{-1}$

(ii) $260\,\text{rev min}^{-1}$.

Hence determine the percentage increase in stored energy for this speed increase. [4]

(e) Assuming operation at 260 rev min^{-1} and two flywheels, estimate the useful electrical energy delivered to the pulse system after accounting for the 78% conversion efficiency. [3]

(f) Compare your answer to (e) with your dissipated-plasma-energy estimate from (c), and comment on whether the energy budget appears physically reasonable. State two factors (other than plasma resistance) that may account for any difference. [4]

(g) Estimate the minimum recharge time from depleted flywheels back to full speed at 260 rev min^{-1} using the given motor power. [3]

(h) A proposal suggests doubling the capacitor-bank voltage while halving total capacitance so that stored energy is unchanged. Discuss one advantage and one disadvantage of this proposal for Tokamak pulse operation. [3]

(i) Open-ended: Propose and justify one modification to the pulse-current profile in Table 2 that could improve overall fusion performance. Your discussion should include a trade-off between plasma heating effectiveness and engineering limits. Use quantitative references from the data. [7]

Total: 33 marks

Suggested Answers / Marking Points for Variant 2

(a) Since $E_{\text{rot}} \propto \omega^2$, moderate speed increase gives significant energy increase. Practical limits: material stress, bearing losses, vibration, safety margins.

(b)

$$Q \approx \text{area under } I\text{–}t \text{ graph (trapezoids)}$$

$$\begin{aligned} Q \approx\ & \tfrac{1}{2}(0+5)(4) + \tfrac{1}{2}(5+8.2)(4) + \tfrac{1}{2}(8.2+8.0)(4) \\ & + \tfrac{1}{2}(8.0+6.3)(4) + \tfrac{1}{2}(6.3+3.8)(4) + \tfrac{1}{2}(3.8+1.2)(4) + \tfrac{1}{2}(1.2+0)(4) \\ =\ & 244\,\text{MA s}. \end{aligned}$$

Hence $Q \approx 2.44 \times 10^8$ C.

(c) Compute I^2 at each time point (with I in A), apply trapezoidal integration for $\int I^2 dt$, then multiply by $R = 1.6 \times 10^{-6}\,\Omega$. Equivalent estimate gives

$$E_R \approx 1.6 \times 10^{-6} \times 1.531 \times 10^{15} \approx 2.45 \times 10^9\,\text{J}$$

(accept values near 2.4–2.5 $\times 10^9$ J depending on rounding).

(d) Using $I_{\text{disc}} = \frac{1}{2}mr^2 = 9.75 \times 10^6\,\text{kg m}^2$:

$$\omega_{240} = 2\pi\frac{240}{60} = 25.13\,\text{rad s}^{-1}, \quad E_{240} = \tfrac{1}{2}I\omega^2 \approx 3.08 \times 10^9\,\text{J}$$

$$\omega_{260} = 2\pi\frac{260}{60} = 27.23\,\text{rad s}^{-1}, \quad E_{260} \approx 3.62 \times 10^9\,\text{J}$$

Percentage increase

$$\frac{E_{260} - E_{240}}{E_{240}} \times 100\% \approx 17.4\%.$$

(e) Two flywheels store $\approx 2(3.62 \times 10^9) = 7.24 \times 10^9$ J. Useful delivered energy

$$E_{\text{useful}} = 0.78 \times 7.24 \times 10^9 \approx 5.65 \times 10^9 \text{ J}.$$

(f) Since $E_{\text{useful}} \approx 5.65 \times 10^9$ J and $E_R \approx 2.45 \times 10^9$ J, budget is plausible; remaining energy can go to magnetic-field energy build-up, induced currents in structures, radiation, and conversion losses.

(g)

$$t_{\min} = \frac{E_{\text{recharge}}}{P_{\text{motor}}} = \frac{7.24 \times 10^9}{18.0 \times 10^6} \approx 4.0 \times 10^2 \text{ s} \approx 6.7 \text{ min}.$$

(h) Advantage: higher voltage can reduce current for same power transfer, lowering resistive cable losses. Disadvantage: greater insulation and switching stress, higher breakdown risk and safety constraints.

(i) Credit justified proposals such as extending high-current plateau or reducing decay gradient; must reference values from Table 2 and discuss trade-offs (e.g., higher heating versus stress, losses, recharge time, component lifetime).